Cell Structure

The cell is the smallest unit of life that can carry out all the processes needed to stay alive. Robert Hooke first described cells in 1665 after looking at thin slices of cork through a simple microscope. Cell theory states that all living things are made of cells and that new cells come only from existing cells.

Prokaryotic cells, such as bacteria, have no nucleus and keep their DNA in a region called the nucleoid. Eukaryotic cells have a true nucleus surrounded by a double membrane called the nuclear envelope. Eukaryotic cells are usually ten to one hundred times larger than prokaryotic cells.

The Plasma Membrane

The plasma membrane is a thin barrier that separates the inside of the cell from its surroundings. It is built from a phospholipid bilayer, with hydrophilic heads facing the water and hydrophobic tails facing each other. Membrane proteins embedded in the bilayer act as channels, pumps, and receptors.

Diffusion is the movement of molecules from an area of high concentration to an area of low concentration. Osmosis is the diffusion of water across a selectively permeable membrane. Facilitated diffusion uses channel proteins to move molecules down their gradient without spending energy. Active transport moves molecules against their concentration gradient and requires energy from ATP. The sodium-potassium pump is a classic example of active transport in animal cells.

Organelles

The nucleus stores the cell's DNA and controls gene expression. Inside the nucleus, the nucleolus assembles the subunits of ribosomes. Ribosomes read messenger RNA and join amino acids into proteins. The rough endoplasmic reticulum is covered in ribosomes and folds proteins that will be exported from the cell. The smooth endoplasmic reticulum makes lipids and breaks down toxins in liver cells.

The Golgi apparatus modifies, sorts, and packages proteins into vesicles for transport. Lysosomes contain digestive enzymes that break down worn-out organelles and food particles. Vacuoles store water and nutrients, and plant cells usually have one large central vacuole that keeps the cell firm.

The mitochondrion is the site of cellular respiration and produces most of the cell's ATP. Mitochondria have their own circular DNA, which supports the endosymbiotic theory. Chloroplasts capture light energy and carry out photosynthesis in plant cells and algae. Chloroplasts contain the green pigment chlorophyll inside stacks of thylakoids.

The cytoskeleton is a network of protein fibers that gives the cell its shape and helps it move. Microtubules form the spindle that pulls chromosomes apart during cell division. Plant cells also have a rigid cell wall made of cellulose outside the plasma membrane.

Cellular Respiration

Cellular respiration releases energy stored in glucose and stores it in ATP. Glycolysis splits one glucose molecule into two pyruvate molecules in the cytoplasm. The Krebs cycle takes place in the mitochondrial matrix and releases carbon dioxide. The electron transport chain on the inner mitochondrial membrane produces most of the ATP. Oxygen is the final electron acceptor in the electron transport chain and combines with hydrogen to form water. When oxygen is missing, cells rely on fermentation, which produces far less ATP than aerobic respiration.

Photosynthesis

Photosynthesis converts light energy, water, and carbon dioxide into glucose and oxygen. The light-dependent reactions take place in the thylakoid membranes and split water molecules. The Calvin cycle takes place in the stroma and uses ATP and NADPH to fix carbon dioxide into sugar. The enzyme rubisco attaches carbon dioxide to a five-carbon sugar during the Calvin cycle.

Cell Division

Mitosis produces two identical daughter cells, each with the same number of chromosomes as the parent cell. The stages of mitosis are prophase, metaphase, anaphase, and telophase. During metaphase, chromosomes line up along the middle of the cell. During anaphase, sister chromatids are pulled to opposite poles of the cell. Cytokinesis divides the cytoplasm and usually follows telophase.

Meiosis produces four genetically different gametes, each with half the number of chromosomes. Crossing over during prophase I swaps segments between homologous chromosomes and increases genetic variation. Errors in meiosis can lead to nondisjunction, where a gamete receives an extra or missing chromosome.